

Windows Internals, Event Logs, Powershell, and Cyber Project

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 3 hours and 30 mins.

Windows Internals + Process Trees

What happens when a program launches in Windows?

*Firstly, Windows receives a request to execute a file. Windows needs to find the executable file now. It locates the program on disk: (HDD/SSD) and it's usually an .exe file. An example path is C:\Program Files\App\app.exe. Windows checks if the file exists, if we have permission to run it, if it is digitally signed or trusted, and if it is 32-bit or 64-bit. After this, a component called the Windows Loader prepares the program. The loader will read the executable format, allocates memory, loads required DLLs, and sets up the process environment. Windows then creates a **process**. A process is a running instance of a program. Remember: process gets PID, memory space, security token, handle table, threads, and environment variables. Ex: chrome.exe -> PID 5124. Then, Windows gives the program its own virtual address space. Most Windows programs depend on DLLs. Ex: kernel32.dll, user32.dll. Windows maps these DLLs into memory. Then, every process starts with at least one thread. A **thread** is the actual execution path the CPU runs. Windows creates the **Main Thread**. Next, the CPU begins the Fetch -> Decode -> Execute Cycle. The processor fetches instructions from RAM, decodes them, and executes operations. The programs cannot directly access hardware safely. So, they make system calls. Ex: open a file, draw windows, etc. Windows kernel handles these requests. Program continues running then eventually closes; threads are terminated and memory is freed.*

What is the Windows kernel?

The Windows Kernel is the core part of the Windows OS. It acts like the central manager between software and hardware. Programs cannot directly control CPU, RAM, disks, devices, and hardware resources. Therefore, they request services through the kernel. Applications ask the kernel for permission to: use memory, access files, talk to devices, use the CPU, and create processes.

What is the difference between user mode and kernel mode?

User mode is the restricted/safe area where normal applications run. Ex: Chrome, Discord, Valorant, Spotify, Word. Programs in user mode CANNOT directly access hardware, CANNOT freely access memory from other programs, and CANNOT fully control the operating system. They must ask the Windows Kernel for services.

Kernel mode is the highest privilege level in the operating system. The Windows kernel and drivers run here. Kernel mode code: CAN directly access hardware, CAN access all memory, CAN control the CPU, CAN modify system resources. It has essentially unrestricted system access.

Install Process Explorer.

Investigating: chrome/msedge, discord, explorer.exe, and svchost.exe

What is svchost.exe?

It stands for Service Host. It is a core Windows process used to run Windows services in the background. Instead of every service running separately, Windows groups many services inside: svchost.exe. This saves memory, CPU resources, and system overhead. So: svchost.exe = container for Windows Services.

Why are there many svchost.exe processes?

There are many because Windows groups different system services into separate containers for stability, security, and resource management.

- 1. Stability: if one service crashes: only that group crashes, not the entire OS.*
- 2. Security Isolation: different services have different permission levels. Some services need network access, system privileges, and user permissions. Windows will separate them to reduce risk.*
- 3. Better Resource Management: Windows can allocate memory separately, monitor CPU usage per group, and restart specific service groups.*

What are parent and child processes?

When one process launches another process: The original process is the Parent Process, and the newly created process is the Child Process.

Find a parent process of browser: check child processes spawned and command line arguments.

My msedge.exe only has msedge.exe child processes. Nothing suspicious!

Imagine Word.exe spawning powershell.exe. This could be suspicious.

Event Viewer + Log Analysis

Windows + R, type **eventvwr.msc**.

Windows constantly records: logins, crashes, services, updates, security events, and system activity.

"What are Windows Event Logs?"

They are built-in records of important activity and events occurring on a Windows system.

"What is the difference between Application, System, and Security logs?"

1. *Application Logs: Record events generated by software applications/programs. These events come from: Chrome, SQL Server, games, Office apps, antivirus software, custom programs, etc.*
 - a. *Ex: Program crashes, Database errors*
2. *System Logs: Record events generated by Windows system components and hardware drivers. These come from: device drivers, services, kernel components, and hardware subsystems.*
 - a. *Ex: Driver failures, Disk errors, Startup/Shutdown, etc.*
3. *Security Logs: Record audited security-related events. These are generated by: Windows Security subsystem, authentication systems, and access control systems.*
 - a. *Ex: Successful/Failed logins, User account creation, etc*

Log Type	Purpose
Application	App events
Security	logins/security
System	OS/hardware

Find: login events, failed events, warnings, service events, and update events.

"What is Event ID 4624?"

A successful account logon occurred.

"What is Event ID 4625?"

An account failed to log on. So, someone tried to log in, but the login FAILED.

Powershell Fundamentals

What is Powershell?

Windows command-line shell and scripting language developed by Microsoft. It is used to: automate tasks, manage systems, run commands, administer Windows, and interact with files, processes, networks, and services. Think of it as a much more advanced version of Command Prompt.

Why do system administrators use Powershell?

It lets them automate, manage, and control Windows systems efficiently. It is better than manually clicking through menus for every task.

In Powershell: Run **Get-ComputerInfo**

Now, **Get-Process**.

Sorting as well, **Get-Process | sort CPU -Descending**

Get-Process | sort WorkingSet -Descending

What is WorkingSet memory?

The amount of physical RAM currently being used by a process. Programs may reserve or request LOTS of memory, but not all of it stays physically loaded into RAM all the time. The Working Set is the portion Windows currently keeps in physical memory for fast access.

Next: **Get-NetTCPConnection**

Try: **Get-Process | Export-Csv processes.csv**

Now: **Get-Service**

But, you can get Running Services only. Try: **Get-Service | Where-Object {\$_.Status -eq "Running"}**

"What are PowerShell pipelines?"

They allow you to pass the output of one command directly into another command. The pipeline operator is: |.
